SW Engineering CSC648-848-05 Spring 2023

StudentSources

Team #04 “Fantastic Four”
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Executive Summary

When studying for finals, the usual routine is to start reviewing from the first lecture and then go from there. Sometimes it’s hard to do so because the resources you used throughout the semester aren’t organized well, then you are going back and forth between topics and weeks of lectures. Students will waste countless hours studying just for it to end up not being that effective. We thought a good idea was a website to organize your bookmarks and resources so that it is easier to go back to those resources. Our website, StudentSources, is a website where any resources from students can be uploaded in an organized fashion. The ordered list can then be shared with other students on the website. The list folders could be titled by class name, class number, or professor. This way students can easily find resources to study within just a few clicks, and the time they spend studying will be effective and useful.

StudentSources will have a range of features to make studying easier for students. The main key feature of the website will be being able to organize online resources such as links to any useful resources or videos. Users will be able to put resources in any order they want, resulting in a list of related resources. The website will also accept multiple file formats, such as PDFs, PNGs, and links to MP4s, documents, pictures, videos. Another key feature of StudentSources is the ability to share those lists of resources with other students. This will prove to be useful when studying alone, with a group, or when wanting to share your study strategies with someone else. The last key feature StudentSources has to offer is the ability to add notes onto PDFs to add any additional information you want to save.

StudentSources is an educational platform offering various unique features to help students enhance their knowledge. Sometimes, the resources that students find online might end up using a different method or formula than what the professor teaches. This makes the resource found practically useless, and makes studying feel pointless without the correct study materials. One key feature of StudentSources is its flexibility. The website's structure allows for the organization of resources based on the professor's name, course name, or course abbreviation and number (CSC648 for example). This feature provides an easy and fast way to navigate the website and find what you are looking for. Other features include collaboration tools such as sharing, commenting, direct messaging, and group chats, as well as the ability to add or remove friends. StudentSources encourages collaboration and sharing of knowledge, hence students can search for publicly shared directories to reference another student’s notes and resources. They can also leave comments on publicly shared resources, boosting the spread of information. Students can add other users as their friend and even mark them

as *Favorite* for easy search. The platform also allows users to direct messages to one another. Students can also create a group chat for group study as group study is also one of the effective means to ace classes. Lastly, our team is planning to include a chatbot powered by AI technology that can summarize documents, saving students valuable time, instead of reading a document with 100 pages of notes. AI chatbot can assist students with their queries and provide personalized support.

Overall, StudentSources offers an innovative platform for students to access and organize their course materials, and even find new course materials from other students. In addition, StudentSources offers a comprehensive set of features designed to facilitate learning and collaboration among students. Apart from those features, the website stands out to be a sustainable and environment-friendly space as the website supports paperless notes organization. We highlight the convenience, collaboration features, and environmental benefits in our platform. By engaging with student communities, promoting sustainability, and offering personalized and collaborative features, we aim to establish our platform as the top choice for students to organize their study materials while contributing to a greener future.

Main Use Cases

1. Struggling with grades (Aaron, 19, 2nd year in college)

Aaron is from Utah but attends SFSU and lives in the dorms. He wanted to go to SFSU to experience living in SF and to meet new people. He is an avid procrastinator and struggles to study for exams and midterms. Recently, he has been struggling grade-wise, so much so that his parents have threatened to send him back to go to college back home in Utah if he doesn't improve his grades. His main problem is that when finals come around, he keeps playing video games and then studies at the last minute. He also needs help staying motivated because he gets frustrated quickly and gives up on studying after a while. He saw someone sitting beside him studying and asked them what website they were using. The student said the website was called StudentSources, and that they made a list of study resources on it. After using StudentSources, Aaron has resources posted by students in SFSU for his Computer Science and History classes. Because of the ordered list of resources, he can easily stay on track and focused. Aaron was also surprised to see that there is a chatbot he could use, especially since everyone has been talking about AI. The chatbot makes his study time more efficient and less stressful. He gets good grades on his finals and finishes the semester with A's and B's, which impresses his parents. His parents let him stay at SFSU, and he is happy he found StudentSources and gets to stay in San Francisco.

2. About to graduate (Sydney, 23, 4th-year college)

Sydney is a senior student from Tennessee who attends SFSU and is excited to graduate this spring. She always wanted to experience living in a big city, and SFSU has provided her with that opportunity. However, Sydney has been feeling overwhelmed lately, as she is away from home and has many final projects and assignments due at the end of the semester. She has always been a procrastinator, making completing her work on time challenging. Sydney is worried that she may not graduate with the high GPA she had hoped for. She mentioned her struggles to a friend, who recommended using a website called "StudentSources," which is a website where students can upload resources like slides and notes to share with other students. After using StudentSources, Sydney found valuable resources from fellow students for her classes to help her stay on track with her assignments. She has found that she can complete her work on time, stay motivated, and get better grades. With the help of

StudentSources, Sydney is confident that she will graduate with the high GPA she has always wanted.

3. A friend needs help with studying (Darren, 22, 3rd-year college)

Darren is a student at SFSU and is very passionate about his studies. His friend, who attends SJSU, has been struggling with their classes lately and has contacted Darren for help. They are both Computer Science majors, so his friend asks him if he has any tips. Darren wants to send his friend a list of study resources to help them improve their grades. He tells his friend he uses a website called "StudentSources," which has a vast collection of resources for various subjects. Darren uses this site to study, so he sends his friend a link to his list of study resources. The resources are well-organized and easily accessible, which makes it easier for his friend to stay focused and motivated while studying. Darren also creates a group chat on the website with his friend and some of Darren's fellow Computer Science classmates. This way they can create a study group together and hopefully be more motivated to study for tests. With the help of the resources, Darren's friend can improve their grades and pass their classes. Darren is happy to have been able to help his friend; he and his friend continue to use StudentSources to stay on top of their studies.

4. Helping students (Dave, 42, Professor of Biology)

Dave is a professor of Biology at CCSF. He enjoys teaching his students and wants to ensure they have all the necessary resources to succeed in his class. One day, he hears about a website called StudentSources that allows him to create a list of study resources that his students can access. He makes a list on StudentSources for his class, including helpful slides, textbook pages, and practice exam PDFs. Dave shares this list with his students at the beginning of the semester, and they find it incredibly useful. They appreciate how organized and easy to use the website is and use it to stay on track throughout the semester. Some students also find other valuable notes uploaded by students for their other classes. Thanks to Dave's uploaded list of resources and the help of StudentSources, his students perform well in his class and feel confident in their understanding of the subject matter. Dave is happy to have found such a helpful tool and looks forward to using it for future classes.

5. Help for tutoring (Milena, 32, Tutor)

Milena is a dedicated tutor who wants to help her students succeed. She tutors middle schoolers mainly with subjects related to math. She has noticed that some of her students need help finding the right study resources to help them with their coursework. To make things easier for her students, Milena has created a list of study resources using StudentSources that she can share with them. She understands that each student has a unique learning style and needs, so she has made sure to include a variety of resources that cater to different types of learners, which is made possible by the StudentSources website. She also has assigned group chats for students so that the students that are doing well in the class can chat with and hopefully help those falling behind. Milena's list comprises online resources, study guides, and practice problems specifically tailored to the subjects her students are studying. Her students are grateful for her help and find the resources helpful in improving their grades. Thanks to Milena's help and StudentSources, her students can achieve academic success and feel more confident in their learning abilities.

6. Technology Troubles (Marc, 20, organized student)

Marc is a scholar student at UC Berkeley who has been acing his class since the first semester. He has been attending classes regularly and making detailed notes of each lecture; moreover, every day at night, he regularly reviews his class notes and researches more on particular topics to gain valuable insights. All this work makes it easy for him to ace the finals since he has to revise his notes. However, two weeks before finals, his laptop fell down the table accidentally and broke along with all of his notes for the current and previous semesters. He went quickly to the service center for repair; however, they said it was impossible to fetch the data. Due to this, he could not study well or perform in the exams. The following semester, he came across StudentSources, which he used to store all his notes. He was happy to organize different links to various videos and articles related to his studies. This helped him stay focused and made him tension free from the laptop crash incident again since he would have a backup of all of his notes.

7. Attending many classes (Zander, 21, college student)

Zander is in a hurry to finish his graduation and has decided to take up all the remaining classes this semester only. He takes six courses for the semester, all of which are study intensive. After two weeks, he realizes he cannot keep up with the assignments and

classes because the subjects and content must be clarified. It is easy to focus in class, but when he goes home and tries to study or work on assignments, he finds it hard to manage different notes and content. He comes across StudentSources, where he can create a list and content categorized by various factors like class name, professor, Etc. He finds it very straightforward to work on it and can manage his assignments well and on time. Moreover, over the semester break, he also adds data from his previous semester. Whenever he is stuck on any advanced course, he can quickly check on the content of the related beginner courses and promptly clear his doubts. Hence, using StudentCourses, he handled the workload very effectively.

8. Landing A Job (Donovan, 26, recent graduate, looking for jobs)

Donovan recently graduated from SFSU and is now applying for jobs in various companies. However, due to the recession and the competitive market scenario, there are very few jobs in his desired field. This has made him very anxious since he is also pressured to repay education loan EMIs. While looking for jobs, he finds out that there are many openings for cybersecurity professionals. The problem is that Donovan last learned about cyber security in his bachelor's. Still, the good news is he used StudentCourses throughout his education to make notes and organize lists of the content. Searching through the course name, he found all the letters and content he used to learn cybersecurity. He was able to re-learn very quickly and crack a job interview as a cybersecurity specialist. This made him very happy and relieved as he could repay his education loan.

9. Overwhelming Course Load Candace — (26, overwhelming course load)

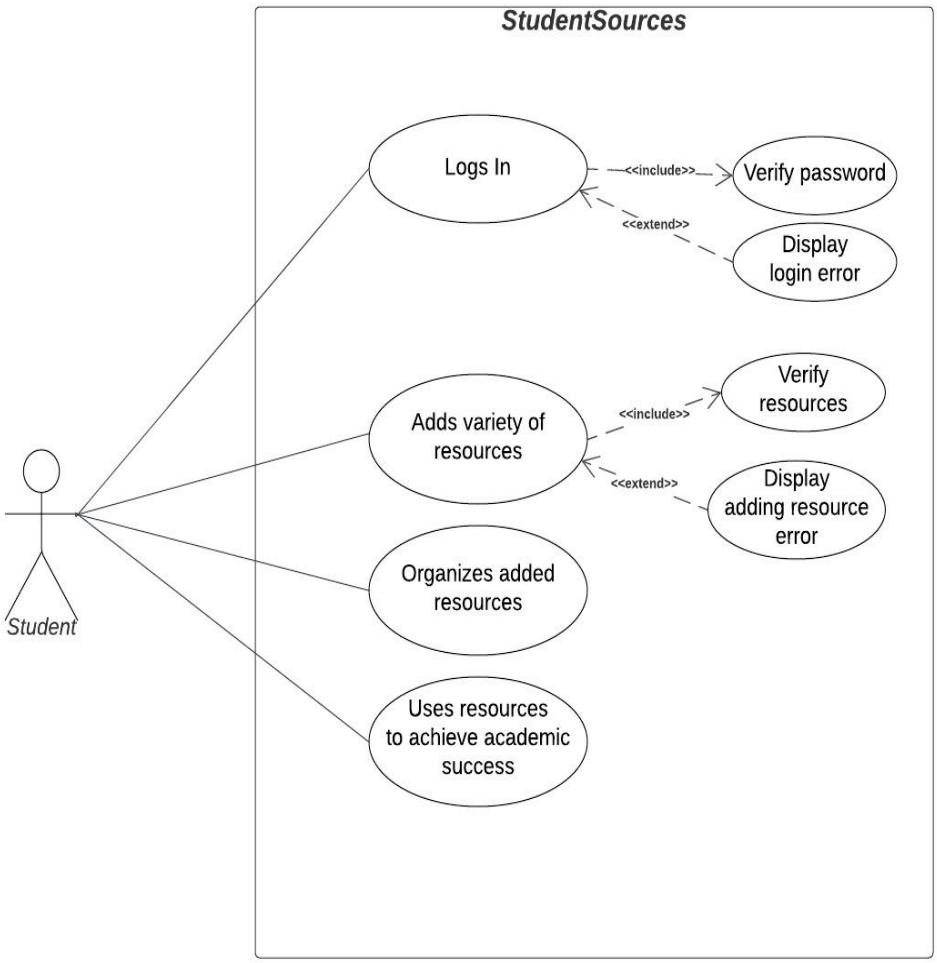
Candace is a Nursing major that has just transferred to SFSU from a community college and realized that the workload at the university level is much higher and finds it overwhelming. The semester begins, and she struggles to keep up with her classes. She considered withdrawing from three of her courses. She then met a classmate taking 24 units, and to her surprise, she "asked how she stays on task with her work?" She responds, "StudentSources, a website that helps organize a list of student resources." They converse more and start up a group chat in StudentSources. Candace and the classmate tell others to join StudentSources if they would like to be a part of the group chat and study together. Candace began using StudentSources and was able to catch up and ended up getting a 3.7 GPA in her first semester as a transfer, and also made some new Nursing major friends along the way.

10. Imposter Syndrome (20, considering dropping out or changing majors)

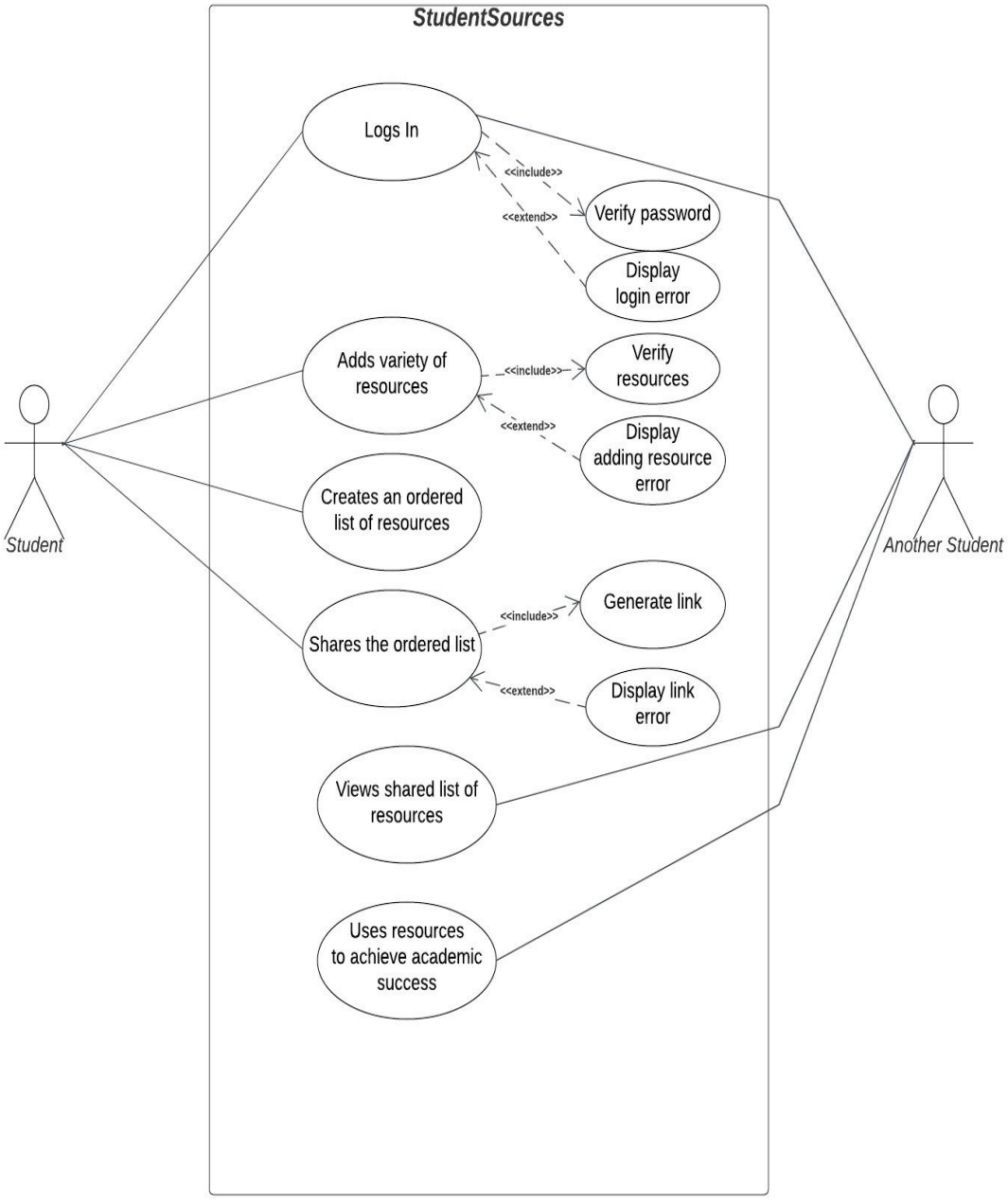
Jacob is 20 yrs old and is a transfer student from Austin, Texas. He is in his first semester and has difficulty learning and feels that the lectures are going too fast for him. He was not able to record proper notes due to the speed of the professor changing the slides. This resulted in him considering dropping out or changing majors. After speaking with some of his fellow classmates, they introduced him to StudentSources, a web page that gives him access to lists of organized resource links for students to improve performance. He starts using the website a lot, and his confidence in passing the class goes up. Using the features like comments and direct messaging, he lets the people who uploaded resources know that they really helped change how he takes and organizes his notes. Jacob does not need to worry about writing everything down because, with the help of StudentSources, he has learned so much more. Jacob decides to continue with his chosen field and is happy with where he is now.

Main Use Cases Diagram

A student organizes his online resources on StudentSources for their academic success.



A student shares his online resources on StudentSources to other students.



Main Data Items & Entities

1. **General User -** A general user can not view *approved resources*. General users do not need to log in/register.

2. **Registered User -** Registered users are attached to an *account*. A registered user can view *approved resources*. A registered user can upload *resources*. A registered user can open their files on a separate *tab*. A registered user must be logged in and registered.

3. **Account -** An account belongs to a *registered user*. An account stores user-related data such as username, real name, and posted resources.

4. **Admin -** An admin manages the website. An admin can access all data and content and modify the database, and verify resources.

5. **Resources -** Resources are the contents added by the users in their account such as text files, images/video, online contents.

6. **Unapproved Resource -** A resource is a specified data grouping that contains useful information for students. Resources can only be viewed by an *admin*.

7. **Approved Resource -** An approved resource is a *resource* that has been approved by an *admin*. Approved resources can be viewed by *admins* and *registered users*.

8. **Directory and Directory Tree -** A directory is a folder that contains resources inside. A directory tree is the structure to organize the contents in the directory.

9. **Comment -** Comment(s) are written remarks left by registered users on their own resource or other users' resources.

10. **Friend -** Friend(s) are other registered users who are added as 'Friend' by a registered user.

11. **Note -** Note(s) are the record of brief description, and thoughts that registered users can add to their account.

Functional Requirements

1. General User

1.1. A general user shall be able to create one or more accounts.
1.2. A general user shall be able to sign up with a unique email address.
1.3. A general user shall be able to sign up with a non-unique password.
1.4. A general user shall be able to see if all requirements for creating a password are met.
1.5. A general user shall be able to sign up with a non-unique username.
1.6. A general user shall be able to read the terms and conditions.
1.7. A general user shall be able to agree to the terms and conditions.
1.8. A general user shall be able to resend an account verification email.
1.9. A general user shall be provided a user manual covering basic features.

2. Registered User

2.1. A registered user shall have at least one account.
2.2. A registered user shall be able to interact with a skippable tutorial sequence.
2.3. A registered user shall be able to view their account information in the Account page.
2.4. A registered user shall be able to temporarily unhide account password information in the *Account* page. By default, when a registered user visits the *Account* page, the password would be hidden for security and password.
2.5. A registered user shall be able to change their account username.
2.6. A registered user shall be able to change their account email.
2.7. A registered user shall be able to change their account password.
2.8. A registered user shall be able to upload any supported file type to their account's cloud storage.
2.9. A registered user shall be able to cancel an upload that is in progress.
2.10. A registered user shall be able to navigate through files with a directory tree.
2.11. A registered user shall be able to organize files alphabetically through a directory tree.
2.12. A registered user shall be able to view the current directory from a *Files* panel.
2.13. A registered user shall be able to create directories.
2.14. A registered user shall be able to create text files.
2.15. A registered user shall be able to create bookmarks of web URLs.

2.16. A registered user shall be able to pin the resources for easy access in the future.
2.17. A registered user shall be able to view supported files on the website.
2.18. A registered user can view the contents of their files on a *Display* panel.
2.19. A registered user can view the contents of their saved web pages on a *Display* panel.
2.20. A registered user shall be able to change the color theme of the website.
2.21. A registered user shall have a place to type and save a note for each resource.
2.22. A registered user may have an assistant AI chatbot.
2.23. A registered user shall be able to highlight text on pdf files.
2.24. A registered user shall be able to comment on files.
2.25. A registered user shall be able to delete/move resources to a trash bin.
2.26. A registered user shall be able to delete resources from a trash bin permanently.
2.27. A registered user shall be able to recover items from a trash bin.
2.28. A registered user shall be able to download files onto their computer.
2.29. A registered user shall be able to share their directories via generated link.
2.30. A registered user shall be able to share their directories by configuring the list as ‘public’.
2.31. A registered user shall be able to configure permissions of their shared directories/folders. The permission allows other users to either only view the list or download the list.
2.32. A registered user shall be able to view other directories shared via link.
2.33. A registered user shall be able to view other directories shared publicly.
2.34. A registered user shall be able to download other shared directories into their account.
2.35. A registered user shall be able to download other shared directories onto their computer.
2.36. A registered user shall be able to search public directories.
2.37. A registered user shall be banned for uploading one or many inappropriate resources by admin.
2.38. A registered user shall not be able to view unapproved resources.
2.39. A registered user shall not be able to share unapproved resources.
2.40. A registered user shall be able to add other registered users as friends.
2.41. A registered user shall be able to remove other registered users from friend-list.
2.42. A registered user shall be able to mark other registered users as their favorite friend.
2.43. A registered user shall be able to edit their posted comments at any time.

2.44. A registered user shall be able to edit their saved notes at any time.
2.45. A registered user shall be able to configure visibility of their resources at any time. The visibility can be either public or private.
2.46. A registered user shall be able to message another registered user.
2.47. A registered user shall be able to create a group chat to message multiple other registered users.
2.48. A registered user shall be able to search other registered users whose profile is set public.
2.49. A registered user shall be able to search friends.
2.50. A registered user shall be able to search pinned resources.
2.51. A registered user shall be able to clear notes added to the resource.
2.52. A registered user shall be able to clear comments added to the resource.
2.53. A registered user shall be able to disable comments in the resource.
2.54. A registered user shall be able to delete a group chat they created.
2.55. A registered user shall be able to rename their resources.

3. Account

3.1. Account shall have a certain amount of cloud storage.
3.2. Accounts shall require and store a username that is at least 6 characters.
3.3. Accounts shall require and store a real name tied to them, and the real name must have a first and last name.
3.4. Accounts shall require and store a password tied to them.
3.5. Accounts password shall be encrypted when stored in a database.
3.6. Accounts shall be created by one or more general users.
3.7. An account shall be owned by one registered user.

4. Admin

4.1. Admin shall be able to access all resources, approved or not
4.2. Admin shall be able to delete any resources present on the website
4.3. Admin shall be able to change an unapproved resource to an approved resource.
4.4. Admin shall be able to change approved resources to unapproved resources.
4.5. Admin shall be able to delete/ban any user for any reason (breaking of rules/TOS, inappropriate username)
4.6. Admin shall be able to delete any inappropriate resources uploaded.
4.7. Admin shall be able to view zero or many unapproved resources.
4.8. Admin shall check if comments on resources are appropriate.
4.9. Admin shall delete zero or many inappropriate comments on one or many resources.

5. Resources

5.1. Resources shall be uploaded by a registered user.
5.2. Resources shall be organized in an alphabetical order.
5.3. Resources such as text files shall be able to be created by a registered user.
5.4. Resources shall be viewed by a registered user.
5.5. Resources shall be commented by registered users.
5.6. Resources shall be moved by a registered user to a trash bin.
5.7. Resources shall be able to be deleted permanently by a registered user from a trash bin.
5.8. Resources shall be recovered by a registered user from a trash bin.
5.9. Resources shall be downloaded on the computer by a registered user.
5.10. Resources shall have comments from other registered users.
5.11. Resources shall be able to be accessed by admin.
5.12. Resources shall be able to be deleted by admin.
5.13. Resources shall be deleted if deemed inappropriate by the admin.
5.14. Resources such as pdf files shall be highlighted by registered users.
5.15. Resources shall have comments verified by admin.
5.16. Resources shall be pinned by a registered user.
5.17. A resource shall have a note created by a registered user.
5.18. Resources shall have configured visibility by a registered user.
5.19. A pinned resource shall be searched by registered users.
5.20. Resources shall be able to be renamed by registered users who added them.

6. Unapproved Resource

6.1. An unapproved resource shall only be viewed by an admin.
6.2. An unapproved resource shall not be viewable by users until approved.
6.3. An unapproved resource shall be able to be converted to an approved resource by admin.
6.4. Unapproved resources shall not be able to be shared until an admin approves it.

7. Approved Resource

7.1. Approved resources shall be able to be viewed by admins and users.
7.2. Approved resources shall be able to be converted back to unapproved resources by the admin.
7.3. Approved resources shall be able to be shared via a generated link.

8. Directory and Directory Tree

8.1. Directory trees shall be navigated by a registered user.
8.2. Directory tree shall have files organized by a registered user.
8.3. Directories shall be created by a registered user.
8.4. Directories shall be able to be shared by registered users to other registered users via generated link.
8.5. Directories shall be able to be shared by registered users by configuring them to be ‘public.’
8.6. Directories shall have configured permissions added by a registered user.
8.7. Directories shared via generated links shall be able to be viewed by a registered user.
8.8. Directories shared by configuring it to ‘public’ shall be able to be viewed by a registered user.
8.9. Directories shall be able to be downloaded by other registered users onto their own account.
8.10. Directories shall be able to be downloaded by other registered users onto their own computer.
8.11. Directories of other users registered configured as public shall be able to be searched by registered users.

9. Comment

9.1. Comments shall be added on resources by registered users.
9.2. Comments on resources shall be verified by admin.
9.3. Comments shall be deleted by the admin, if deemed inappropriate.
9.4. Comments shall be able to be edited at any time by the registered user who posted it.
9.5. Comments shall be able to be cleared by a registered user at any time.
9.6. Comments shall be disabled by a registered user at any time.

10. Friend

10.1. A friend shall be added by registered users.
10.2. A friend shall be removed by registered users.
10.3. A friend shall be added as a favorite by registered users.
10.4. A friend shall be searched by registered users.

11. Note

11.1. A note shall be typed and saved by a registered user for each resource.
11.2. A note shall be able to be edited at any time by the registered user that saved it.
11.3. Notes shall be cleared by registered users at any time.

Non-Functional Requirements

1. System Requirements

1.1. System shall always hide registered users account password information on startup the *Account* page.
1.2. System shall periodically delete files from a trash bin.
1.3. System shall provide information to registered users the amount of free storage remaining.
1.4. System shall display an error message when a registered user exceeds the cloud storage limit for their account.
1.5. System shall show a registered user a progress and estimated wait time upon uploading files.

2. Performance

2.1. Any response to StudentSources links shall be within 2 seconds.
2.2. Any response to any StudentSources links shall be within 4 seconds during high volume
2.3. The website shall be able to load text and image files with large sizes within 3 seconds and audio and video with large sizes within 10-15 seconds.
2.4. The website shall be able to load text and image files with large sizes within 5 seconds and audio and video with large sizes within 20 seconds during high volume.
2.5. Log-in and logout of websites shall be almost instant.
2.6. Buttons, textboxes, and options on the landing UI shall be responsive.
2.7. Buttons, textboxes, and options on upload content UI shall be responsive.
2.8. StudentSources shall be able to handle a large number of simultaneously active users.
2.9. Uploaded audio and linked audio and video files shall be streamed to promote better user experience.

3. Storage

3.1. StudentSources shall be able to handle files with a large size, with the max size being 1 GB.
3.2. The cloud storage container shall properly store files in TXT format.

3.3. The cloud storage container shall properly store files in PDF format.
3.4. The cloud storage container shall properly store files in MP3 format.
3.5. The cloud storage container shall properly store files in URL format.
3.6. The cloud storage container shall properly store files in JPEG format.
3.7. The cloud storage container shall properly store files in JPG format.
3.8. The cloud storage container shall properly store files in PNG format.
3.9. The cloud storage container shall properly store files in GIF format.
3.10. The cloud storage container shall be able to handle different file formats while still being efficient.
3.11. The database and cloud storage shall store metadata associated with files stored in the cloud storage container.

4. Security

4.1. Users shall have a registered account to have access to all StudentSources files.
4.2. Passwords shall be encrypted in the database.
4.3. The website shall use secure hosting (AWS).
4.4. The website shall ensure only registered users can access resources.
4.5. The website shall ensure only admins can access sensitive information.

5. Environmental

5.1. The website shall be optimized for energy efficiency, optimizing images, using efficient code.
5.2. StudentSources shall encourage students to use electronic files instead of printing and using materials to reduce paper waste and the environmental impact of printing.
5.3. StudentSources shall encourage responsible use by prompting users to delete unwanted files when max capacity of the account is reached to reduce storage and bandwidth of the website.

6. Marketing

6.1. StudentSources shall be marketed primarily to students.
6.2. The website shall be marketed as a better way for students to store and organize study resources.
6.3. The website will be marketed as a way to reduce paper waste.

7. Legal

7.1. StudentSources shall comply with copyright laws, to ensure that when the website stores files without infringing the copyrights of others.

7.2. StudentSources shall obtain permission to use copyrighted material according to fair use and copyright laws.
7.3. StudentSources shall have a privacy policy explaining how personal data is collected, used, and protected, which is important for things like student information.
7.4. StudentSources shall have security measures to prevent unauthorized access of data stored in the website's database.
7.5. StudentSources shall have terms and conditions that govern the use of the website by users.
7.6. Terms and Conditions shall explain rules about uploading and downloading files and acceptable use of the site.

8. Content

8.1. Content present on the website shall be mostly if not all related to academic courses.

9. Look and Feel

9.1. StudentSources' landing page UI shall be simple for users to understand.
9.2. The website shall be easy to navigate, with users only spending no more than 3-5 minutes browsing around.
9.3. StudentSources UI shall utilize non-distracting colors that complement the content.
9.4. StudentSources UI for uploading content shall be easy to understand and navigate.
9.5. The website shall have a clean and minimalist design,avoiding clutter and aiming for a sleek appearance.

10. Usability

10.1. The website shall have no issues running when experiencing high volume.
10.2. The website shall maximize user experience in storing and organizing study resources.

11. Privacy

11.1. StudentSources shall only collect data that is necessary for the website's purpose.
11.2. Passwords shall be encrypted to ensure data security.

12. Compatibility

12.1. The website shall be accessible using branded computers such as Windows, Apple, Asus, etc.

12.2. The website shall be accessible on a range of devices like desktop computers, laptops, tablets, smartphones.
12.3. Website shall be compatible with operating systems such as Windows and macOS.
12.4. Website shall be compatible with a wide range of web browsers, including but not limited to: Chrome, Firefox, Edge, Safari.
12.5. The website shall support uploading of a range of file formats, like TXT, PDF, MP3, URL, JPEG, JPG, PNG, GIF.

13. Organizational

13.1. StudentSources shall have a user-centered design, focused on ensuring the website is easy to use and navigate for users.
13.2. The website shall have a clear purpose for students and stakeholders and not be too broad.

14. Coding Standards

14.1. The source code shall be clear.
14.2. Source code shall be concise.
14.3. Source code shall be consistent.
14.4. Source code shall be well commented.
14.5. Source code file names shall be clear.
14.6. Source code file names shall be concise.
14.7. Source code file names shall be consistent.
14.8. The file structure of source code shall be easy to navigate.

Competitive Analysis

Important features of competitive products

Company Features	Padlet	Wakelet	Bublup	LiveBinders	Dropbox
Strengths	Customizable and collaborative.	Intuitive UI placement. Offers features for students of all levels	Lots of functionality	Good for sharing docs in a binder format.	Very fast upload and syncing.
Weaknesses	Unfamiliar user interface and limited cloud storage. Too costly for students.	Limited features.	Some features do not work as expected. Offers low storage.	Difficult for the average user. Storage is limited for all levels/tiers	Lacks themes, customization, and easy sharing.
Pricing	Free tier, $6.99/ month tier, and $9.99/ month tier.	Free tier, then requires a quote for upgrading.	Free tier, $6.99/ month tier, $9.99/ month tier, and $24.99/ month tier	Free tier, and $19.95/ month paid tier.	Provides freemium service with several plans. $9.99/ month - $24/ month plans
Social Media	Instagram.	Facebook, LinkedIn, Twitter, Instagram	Facebook, Pinterest, Instagram, YouTube, Reddit, TikTok.	Blogs, podcasts, YouTube, Twitter, Facebook, LinkedIn, and Instagram.	Twitter, Facebook, and YouTube.
Experience	Moderate learning curve with unfamiliar UI.	Easy to navigate, set up, and organize resources.	Provides plenty of instructions throughout the setup process.	Features are confusing to use.	Provides plenty of support and features in an intuitive manner.

Key Features of Competitors vs. StudentSources

Company Features	Padlet	Wakelet	Bublup	LiveBinders	Dropbox	StudentSources
UI Customization	+	-	+	+	-	++
Easy User Interface	++	+	+	-	+	++
Cloud Integration	-	+	-	+	+	+
Custom Collection Sharing	-	+	-	+	+	++
Group Collaboration With Comment Feed	+	-	+	-	-	++

+ Feature exists **++ Superior** **- Does not exist**

Competitive Analysis Summary

After conducting our competitive analysis, we can see that the market for storage of general resources is out there but is usually limited and locked behind an expensive paywall, which for a college student is not optimal. With more essential expenses to worry about, such as housing, tuition, and textbooks, students like us will look for more cost-friendly options. Since our target audience is mainly college students, our website is superior due to the website being geared towards students. Competitors such as DropBox or LiveBinders have cloud storage for files but are mostly used for general file storage; in StudentSources, only files related to college and specific classes will be present. This results in students being able to easily find and access valuable resources from students who have taken or are taking same or similar courses. Based on data compiled from other study websites, we have concluded that the website we plan to create, StudentSources, is a better alternative for students based on convenience, UX/UI, and low pricing.

High-level System Architecture and Technologies Used

Server Host: Amazon EC2 Dedicated Host, t2.micro 1vCPU 1 GB RAM
Operating System: Ubuntu 22.04 Server
Database: MySQL 8.0.28
Web Server: NGINX 1.18.0
Server-Side Language: Javascript and Java
Additional Technologies:

- Web Framework: React and Spring Boot
- Web Wireframe: Figma
- IDE: VSCode and IntelliJ
- Web Analytics: Amazon Kinesis Data Analytics
- Runtime Environment: Node.js and Java Runtime Environment
- File-System: Amazon Simple Storage Service (Amazon S3)

List of Team Member Contributions

Team Member	**Contribution**
Saru Shrestha	- Use case diagram - Research competing products - Research SW Stack architecture and tools - Configure Host server - Configure Database server - Revise Milestone 1 documentation
Kilian Kistenbroker	- List functional requirement - Research SW Stack architecture and tools - Research important features of competitive products - Configure Host server - Backend (request/response) program
Elias Abay	- List non-functional requirement - Research key features of competitors vs. StudentSource
Parth Panchal	- Main use cases - Research key features of competitors vs. StudentSource - Manage official GitHub Repository - Set up and manage backup GitHub Repository
Joshua Del Rosario	- Main use cases - List functional requirement - List of main data items and entities - Ordering of non-functional requirement - Competitive Analysis Summary - Format Document
Wilson Yin	- List non-functional requirement - List functional requirement - List of main data items and entities
Omar Thiongane	- Main use cases

	- Use case diagram - Research important features of competitive products - Format Document

Team Member Contribution Score

Team Member	**Score**
Kilian Kistenbroker	10
Elias Abay	10
Parth Panchal	10
Joshua Del Rosario	10
Wilson Yin	10
Omar Thiongane	10

Scale: 1 to 10, where a score of 1 means “no work at all”, and score of 10 means “contributed to the maximum capacity”